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M168_Notes_7.1

# CHAPTER 7: PROBABILITY

## *7.1: Experiments, Sample Spaces and Events*

An ***experiment*** is an activity with an observable result. Tossing coins, rolling dice and choosing cards are all probability experiments.

The result of the experiment is called the ***outcome*** or ***sample point***. The set of all outcomes or sample points is called the ***sample space*** of the experiment.

*Example*
What is the sample space for flipping a fair coin? Rolling a 6-sided die?

$S = \{H, T\}$, $S = \{1, \ldots, 6\}$

An ***event*** is a subset of a sample space. That is, an event can contain one or more outcomes that are in the sample space.

*Example*
What are all possible events for the experiment of flipping a fair coin?

$\phi, \{H\}, \{T\}, \{H, T\} \rightarrow 4 = 2^2$

*Example*
How many events are possible when a six-sided die is rolled?

$2^6$

*Example*
What is the sample space for flipping a fair coin three times?

Find the event E where $E = \{x|x \text{ has exactly one head}\}$

Find the event $E$ where $E = \{x|x \text{ has two or more heads}\}$

Find the event $E$ where $E = \{x|x \text{ has more than 3 heads}\}$

| |
|---|
| A sample space in which each of the outcomes has the same chance of occurring is called a UNIFORM SAMPLE SPACE. |

*Example*
A bowl contains the letters AGGIES. How many outcomes are in the uniform sample space?

What is the uniform sample space for rolling two fair six-sided dice?

| | | | | | |
|---|---|---|---|---|---|
| 1~1 | 2~1 | 3~1 | 4~1 | 5~1 | 6~1 |
| 1~2 | 2~2 | 3~2 | 4~2 | 5~2 | 6~2 |
| 1~3 | 2~3 | 3~3 | 4~3 | 5~3 | 6~3 |
| 1~4 | 2~4 | 3~4 | 4~4 | 5~4 | 6~4 |
| 1~5 | 2~5 | 3~5 | 4~5 | 5~5 | 6~5 |
| 1~6 | 2~6 | 3~6 | 4~6 | 5~6 | 6~6 |

These sample spaces have all been finite. That is, we can list all the elements. An infinite sample space has to be described; you can't list all the elements:

*Example*

What is the sample space for the time spent working on a homework set?

Describe the event of spending between one and two hours on a homework set.